

Nikita Vostrosablin, PhD

Engineering Leader focused on production ML and backend systems. I bring experience from both research-heavy and product-focused environments.

PRAGUE, CZECHIA

CONTACT

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EXPERIENCE

Stats Perform

> Engineering Manager | Jul 2025 - Present | Prague, Czechia

Lead a team of 9-12 engineers owning production ML and data pipelines for sports analytics products.

- Own architecture, reliability, and operational health for ~30 production ML/data pipelines and ~150 distributed Python services across AWS and on-prem Kubernetes, including systems processing thousands of events/sec and 25 Hz streaming workloads.
- Drove architectural decisions and launches for new production pipelines and systems, while improving the reliability and operability of existing ones.
- Defined pipeline SLOs and introduced structured observability, alerting, and on-call practices, cutting incident resolution from multiple days to ~1 day and support load from ~25% to ~2% of team capacity.
- Improved key pipeline processing times by up to ~80% through architectural redesign and algorithmic optimization (*example*: by removing Kafka single-topic bottlenecks with fan-out service patterns).
- Established service standardization, end-to-end testing, and integration testing, enabling multiple deployments per day and reducing idea-to-production lead time to ~1 week.
- Hired 3 engineers, led 3 promotions, established the DX4 framework in the team, and reorganized the team into a squad-based ownership model to improve focus and accountability.
- Initiated a reliability, quality, and developer experience program to reduce technical debt and standardize services, and drove AI adoption to workflows where ~75-80% of code is generated by agents under controlled review.

> Senior Software Engineer, Team Lead | Dec 2023 - Jul 2025 | Prague, Czechia

Acted as technical lead for a 4-engineer sub-team responsible for productionizing ML pipelines and distributed systems for the Opta Vision platform, delivering AI-driven metrics across multiple competitions.

- Designed and implemented event-driven pipelines and services for real-time and post-game metric generation, and rebuilt underperforming flows to improve throughput.
 - Led migration of compute workloads from AWS-managed services to Kubernetes as part of a broader cost-optimization effort.
 - Drove architecture decisions, introduced design-first development through design docs, mentored engineers, and contributed to hiring.
 - *Example of impact*: redesigned DynamoDB access patterns and schema to eliminate expensive scan operations, improving performance while reducing cost by up to 85%.
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Merck (MSD Czech Republic)

> Scientific Software Engineer, Team Lead | Jul 2021 - Nov 2023 | Prague, Czechia

Led a small team of 2 contractor software engineers delivering scientific software and platform capabilities for drug discovery.

- Launched a platform for mRNA molecule optimization.
- Participated in open-sourcing part of it as [mRNAid](#), an experimentally validated open-source tool for optimization and visualisation of mRNA molecules.

- Contributed to patented systems for drug discovery, including Reaction Synthesis Pathfinder (Publication No. 20240105285) and Chemistry, Manufacturing, and Controls Change Assessment System (Publication No. 20240004935).
- Built AWS event-driven genomics platforms with Batch, Lambda, Step Functions, SQS, DynamoDB, and Redshift, and implemented Terraform, CI/CD, and production observability.

> **Python Developer / Machine Learning Engineer | May 2019 - Jul 2021 | Prague, Czechia**

Worked across AI and bioinformatics departments on molecular computation and optimization problems for drug discovery.

- Participated in open-sourcing [Mutation Maker](#), an application for mutagenic primer design used in protein engineering workflows.

Dimension Data

> **Data ETL Engineer | Dec 2017 - Apr 2019 | Prague, Czechia**

Built BI infrastructure for a major European client, covering automated dataflows, reporting, and data-quality processes.

ORIFLAME SOFTWARE

> **Intern - Data Scientist, BI/DWH Developer | Jun 2017 - Aug 2017 | Olomouc, Czechia**

Applied ML and NLP methods to sales forecasting and review sentiment analysis, which was later integrated to the sales operations worldwide.

Freelance

> **Freelance Web Developer | Sep 2016 - Dec 2016**

Built web tooling and automated mail reporting for an appartement renovation web service using Node.js, HTML, CSS preprocessors, and template engines.

DTU - Technical University of Denmark

> **Researcher | Sep 2016 - Nov 2016 | Kongens Lyngby, Denmark**

Did a research in quantum optomechanics, focusing on a stroboscopic optomechanical transducer with an experimental team.

EDUCATION

- **Univerzita Palackeho v Olomouci**
PhD, Quantum Physics (Sep 2014 - Mar 2021)
- **Lomonosov Moscow State University (MSU)**
Master of Science, Physics (2008 - 2014)

LANGUAGES

- Russian (Native)
- English (Fluent, Professional Working)
- French (Limited Working)
- Czech (Intermediate)

TECHNOLOGIES

- **Languages & Core:** Python, Go, SQL, Bash, C++, JavaScript
- **Cloud & Infrastructure:** AWS, Kubernetes, IaaS (Terraform), CI/CD (Jenkins, ArgoCD)
- **Streaming & Messaging:** Kafka, Apache Flink, SQS, SNS
- **Databases:** Relational (Redshift, PostgreSQL, MySQL), NoSQL (DynamoDB), Graph (Neo4j)
- **ML / Data Science:** pandas, Polars, scikit-learn, PyTorch
- **Orchestration:** Airflow, Temporal
- **Observability:** ELK, Grafana, OpenTelemetry